Identifying human pre-mRNA cleavage and polyadenylation factors by genome-wide CRISPR screens using a dual fluorescence readthrough reporter

Paper by:

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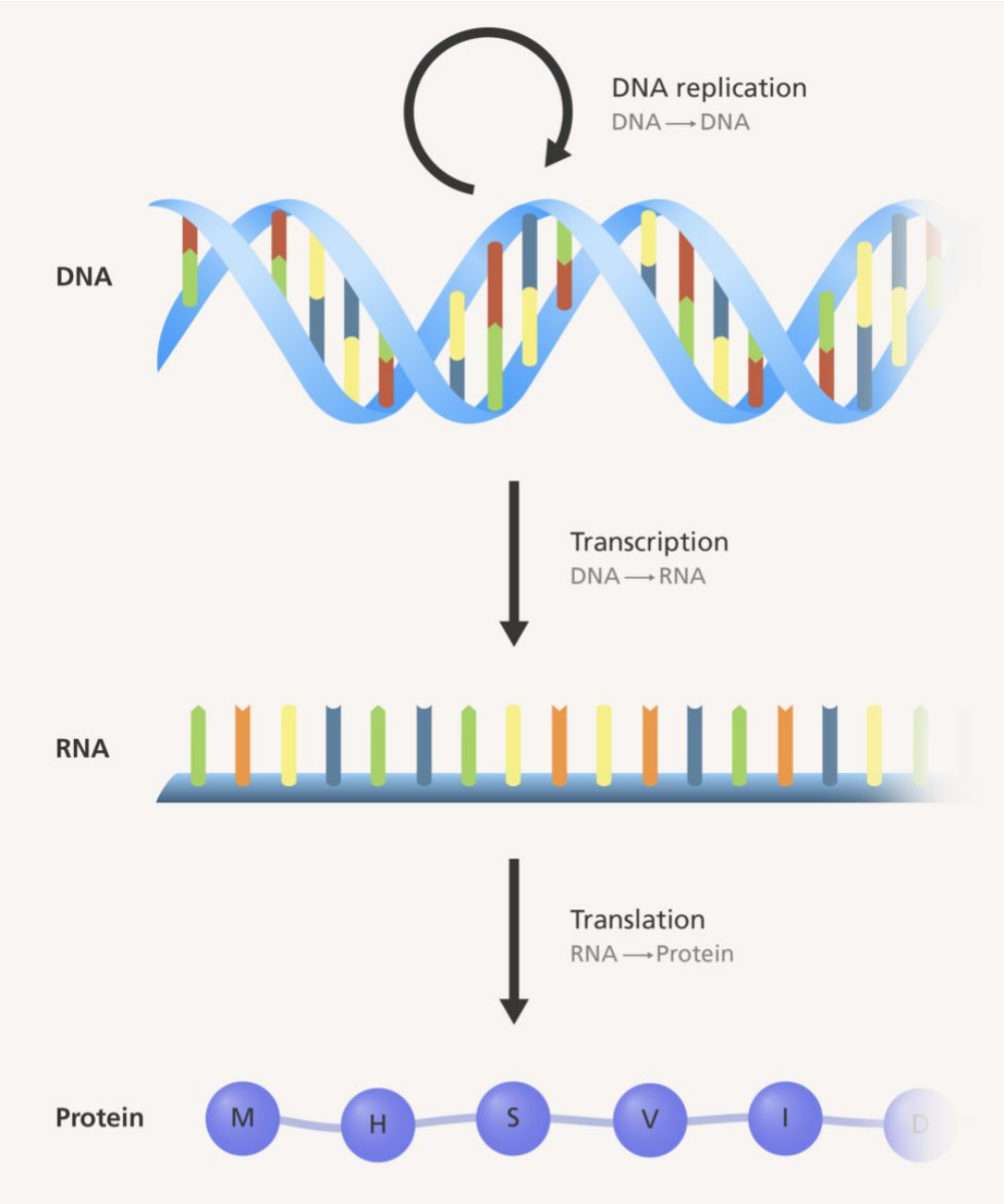
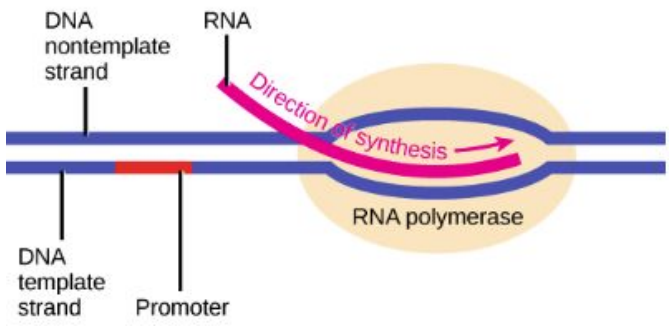
Central dogma of Molecular Biology

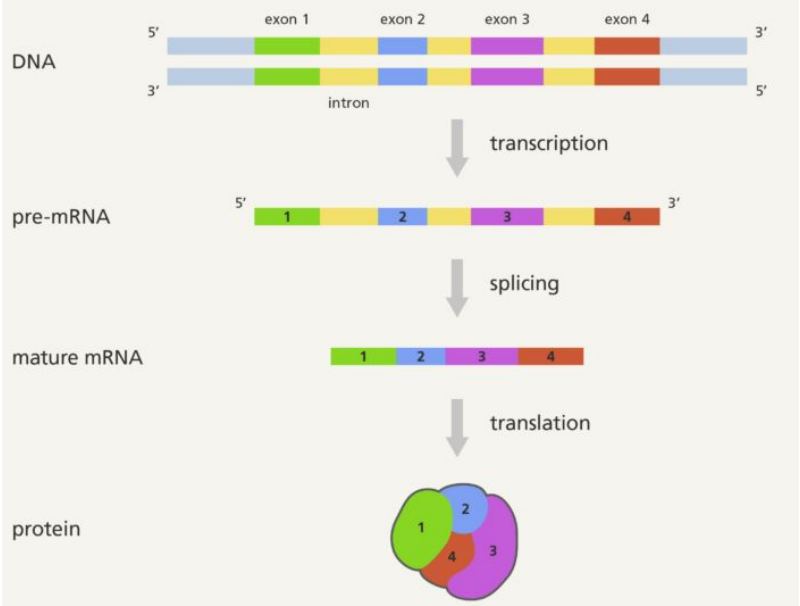
- DNA ▫ RNA ▫ Protein



What is pre-mRNA?

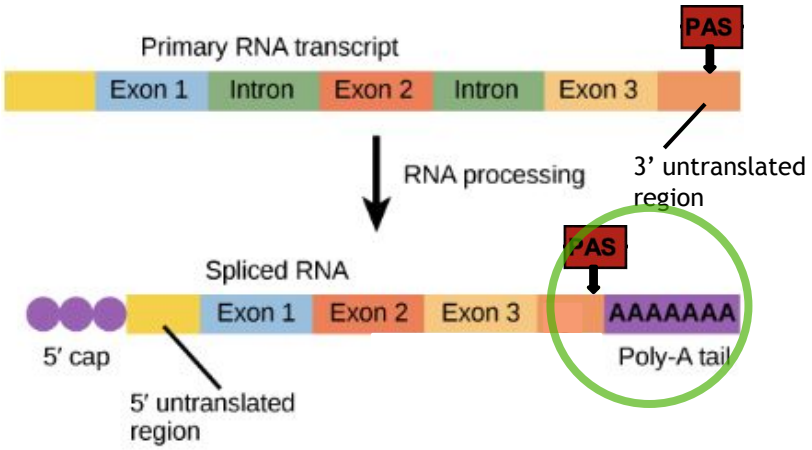


- During transcription, RNA pol 2 binds to the promoter region of a gene and synthesizes a precursor messenger RNA (pre-mRNA) strand complementary to the DNA template.

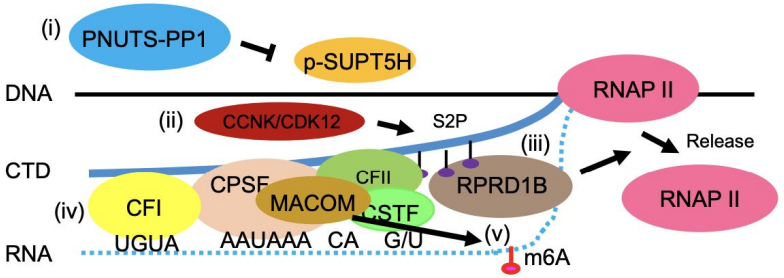


- After transcription, pre-mRNA undergoes various modifications before translation, including (but not limited to):
  - 5’ capping
  - Intron splicing
  - Exon joining
  - Polyadenylation
  - Mature mRNA export

Cleavage and Polyadenylation - CPA



- Polyadenylation can generally be broken down into two steps
  - Cleavage at the PAS site
  - Addition of a Poly (A) tail that provides the mRNA with stability (circled)



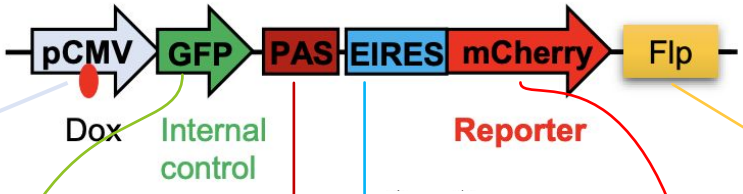
- The actual process is much more complex and implicates many different proteins - CPA factors.

Objective and Hypothesis

- The objective of this study is to develop an effective method for measuring CPA efficiency and prove it's utility by confirming the existence of proteins already known as CPA factors, and identifying possible new ones.
- The study aims to achieve this via
  - Development of a Dual Fluorescence Readthrough Reporter System
  - Conduction of a genome-wide CRISPR Cas9 screen
  - Validation of identified CPA factors (new and old)

- Hypothesis: The Dual Fluorescence Reporter system and CRISPR knockout screen combine to generate fluorescence ratios reflecting 3' end processing efficacy. Differences in these ratios will help identify CPA factors, both old and new.

Dual Fluorescence Reporter



CMV promoter

Inducible by Doxycycline - part of Flp-In T-Rex System

Green fluorescent protein.

Always expressed once integrated.

Red fluorescent protein, only transcribed during transcriptional readthrough.

Signifies inhibition of CPA mechanism.

Flp-In T-Rex System

Allows for stable integration of transgenes in a specific genomic location

Cleavage site for CPA machinery.

This is where the reporter is cut when the CPA mechanism is functioning properly.

EMCV Internal Ribosomal Entry Site

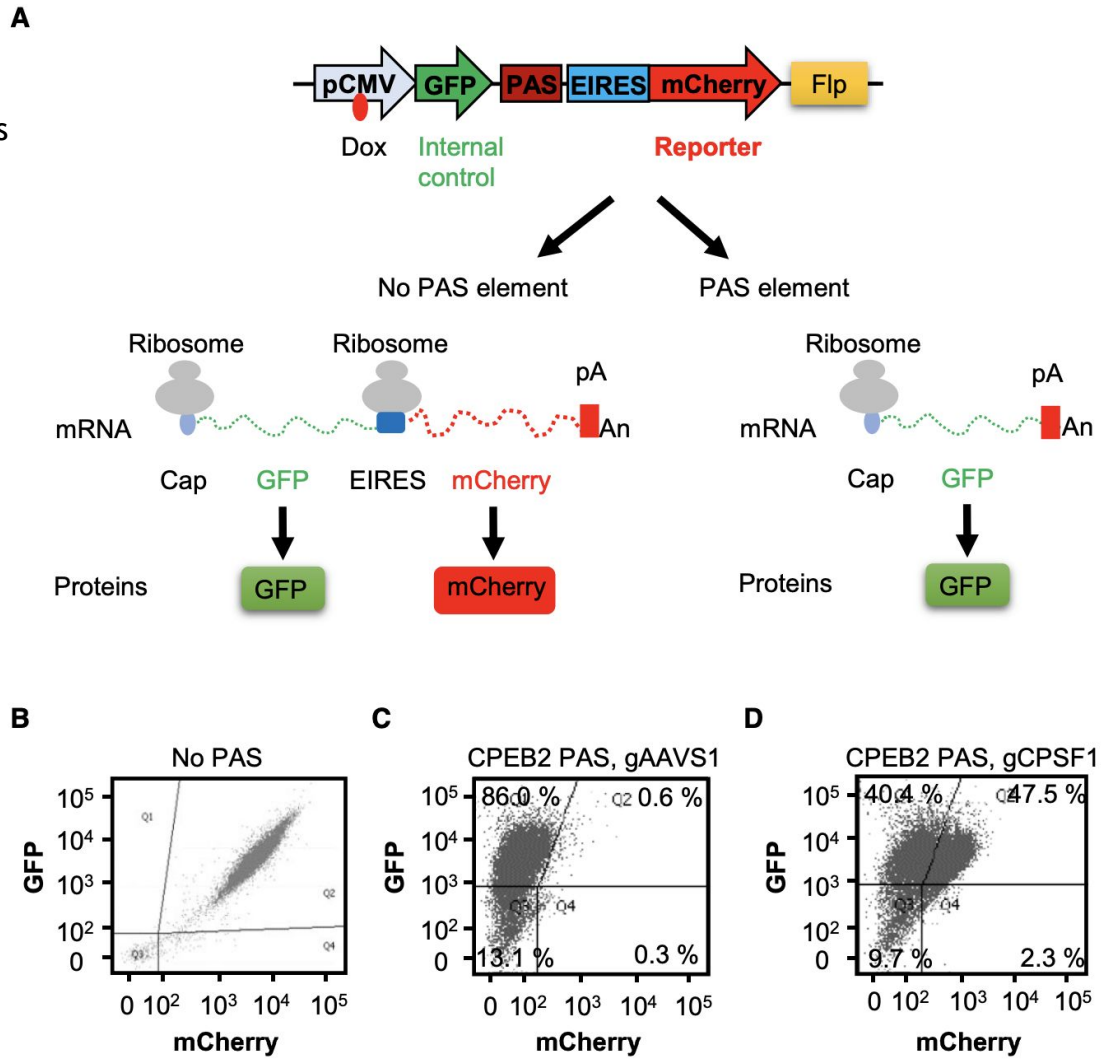
Enables independent translation of mCherry unless cleavage occurs

Experimental Design:

The reporter works is by giving off different fluorescent signals depending on it's structure, specifically it's length.

If the CRISPR screen has knocked out a CAP factor it will inhibit or destabilize the overall CAP mechanism and prevent cleavage.

Transcriptional readthrough will occur, the PAS site will be ignored by RNA pol II, and both FP's will be expressed

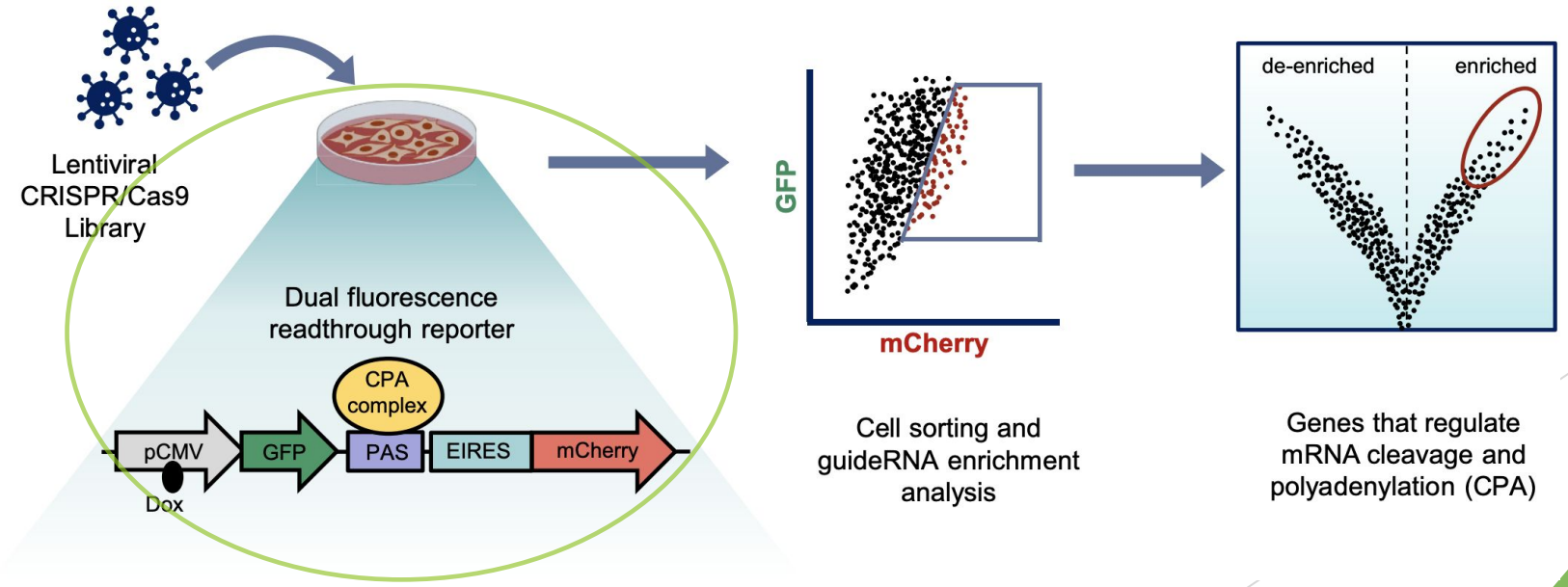


If CAP functions correctly, then the reporter is cleaved at the PAS site by CAP proteins.

A cleaved reporter only contains GFP and gives off green light read by the cytometer.

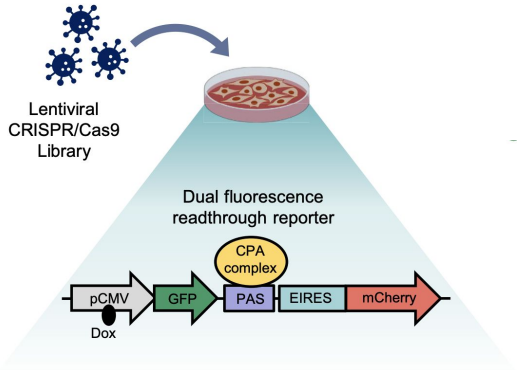
Establishing the parental cell line

- Once the reporter has been cloning, it is transfected into the T-REx HEK293 cell line. An antibiotic called Hygromycin B is then added to the cell culture media for selection.
  - Along with the components mentioned before, the reporter also has a Hygromycin resistance gene. This ensures that any cell that has stably integrated the reporter DNA, is also resistant to Hygromycin.
  - Adding the antibiotic to a dish with cells will kill off any cells that do not have the resistance gene, leaving only those containing the reporter.
  - Fluorescence is induced via Dox and subsequently sorted via FACS, then expanded.



Genome wide CRISPR Cas9 screen

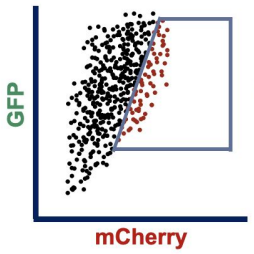
- Once the cell line has undergone stable integration of the reporter, the lentiviral CRISPR library is added.
- A genome wide CRISPR library is a comprehensive collection of guide-RNA's that target protein coding genes (18,053 in this case). These gRNA's are accompanied by a Cas 9 protein that "turns off" or *knocks out* the gene targeted by the gRNA.

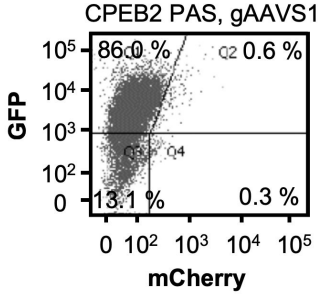


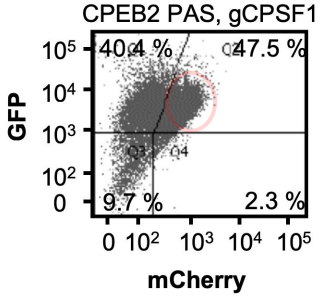
  - The gRNA's and Cas 9 proteins are contained within virus particles that help with entry into the cell
  - This library has 4 different gRNAs targeting each gene to ensure efficacy of the KO

Sorting

- As with the establishment of the parental cell line, after library transduction the cells receive Doxycycline to induce fluorescence, and then are sorted according to certain fluorescence parameters.
- Cells with altered mCherry to GFP ratios are collected by the sorter for further analysis. This process occurs 2x for a purified cell population.
  - Most cells will have a fluorescence profile that favors GFP
  - The cells of interest are the ones with profiles showing increased mCherry expression along with GFP, shown here
- Cells expressing mCherry contain the full reporter structure. This means no cleavage has occurred due to the CPA mechanism being disrupted by the gRNA-Cas 9 protein complex.







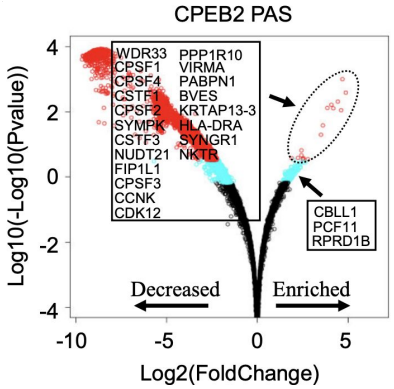
gRNA enrichment

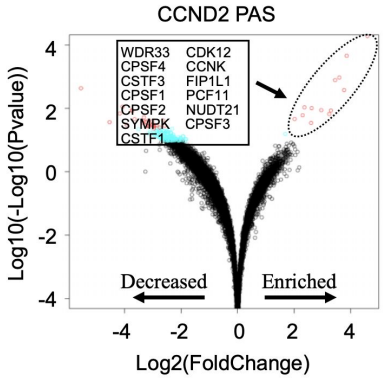
- Once you have a population of cells with their 3’ end processing mechanism disrupted, you need to know which genes were knocked out to cause this phenotype.
- Genomic DNA is extracted from both sorted (high mCherry/GFP ratio) and unsorted populations
  - The regions of the genome containing integrated gRNAs are amplified via PCR
  - The gRNA sequences undergo next-generation sequencing
- The sequencing data generated from both populations is analyzed to determine abundance of each gRNA, and the data is normalized.
  - DESeq2 software used to differentiate abundance levels between the two populations
  - Statistical significance calculated
- When gRNAs targeting a specific gene were enriched in the sorted population (cells with disrupted CPA), that gene is regarded as a potential CPA factor.

Results

- Correctly identified known CPA factors:
  - *CPSF* – Cleavage and Polyadenylation Specificity Facto
  - *CSTF* - Cleavage Stimulation Factor
  - *CFIm* - Cleavage Factor Im
  - *CPSF73*, *CPSF100*, and *Fip1*

- Discovered Novel CPA factors
  - *CDK12* – Kinase responsible for RNA Pol II phosphorylation at CTD S2
  - *CCNK* – Regulator of CDK12
  - *RPRD1B* – Binds RNA and regulates release of RNA pol II at 3' end of genes





Conclusion

- Key findings:
  - Validation of known CPA factors
  - Identification of Novel CPA factors
  - The dual fluorescence reporter and CRISPR screen combination make for an effective assay

- Future directions
  - More investigation into how the novel CPA factors contribute to the CPA process
  - This assay could be applied to different cell types to interrogate situational specific CPA regulation
  - The dual sense reporter could be used for investigating disease models with mRNA processing abnormalities

Critique

- Cons:
  - Context - this paper needs more exposition. It jumps right into a highly technical methods section. More explanation of CPA would be helpful.
  - Methods section
    - The way they presented the promoters and plasmid constructs was difficult to follow. Instead of writing out the DNA sequences for promoters and primers in paragraph form, plasmid maps might have been clearer. Saving the specific sequences for the supplemental data section could have made the paper more succinct.
    - In the places where it is unavoidable, a table would have worked.
- Pros:
  - Successful reporter – in addition to 3' processing studies, this reporter could be utilized in any studying involving transcriptional readthrough.
  - Multidisciplinary and cutting edge – the paper falls uses techniques from a varied and innovative subject areas including synthetic biology, and bioinformatics.

Image credits

- Image slide 2: Laura Olivares Boldú / Wellcome Connecting Science
  - https://www.yourgenome.org/theme/how-is-dna-turned-into-protein-the-central-dogma-of-molecular-biology/
- Image 1 slide 3: Concepts of Biology, Charles Molnar
  - https://press.rebus.community/conceptsofbiology/chapter/9-3-transcription/
- Image 2 slide 3: Laura Olivares Boldú / Wellcome Connecting Science
  - https://www.yourgenome.org/theme/what-is-rna-splicing/
- Image 1 slide 4 (edited by Tahreem): Concepts of Biology, Charles Molnar
  - Same as above
- All other images are from the paper itself.
  - https://academic.oup.com/nar/article/52/8/4483/7642059